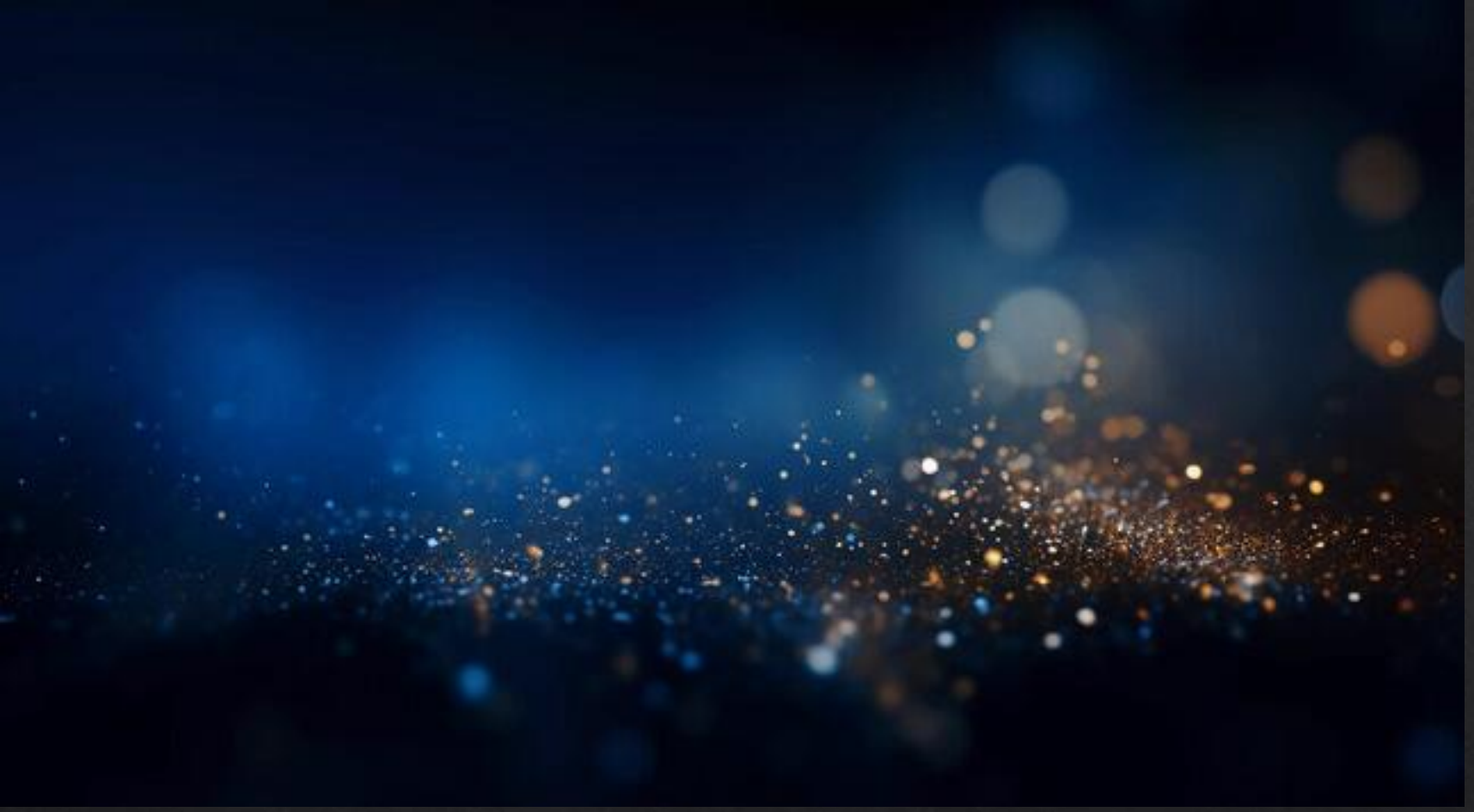


MA416

Lecture 24 – November 25, 2025

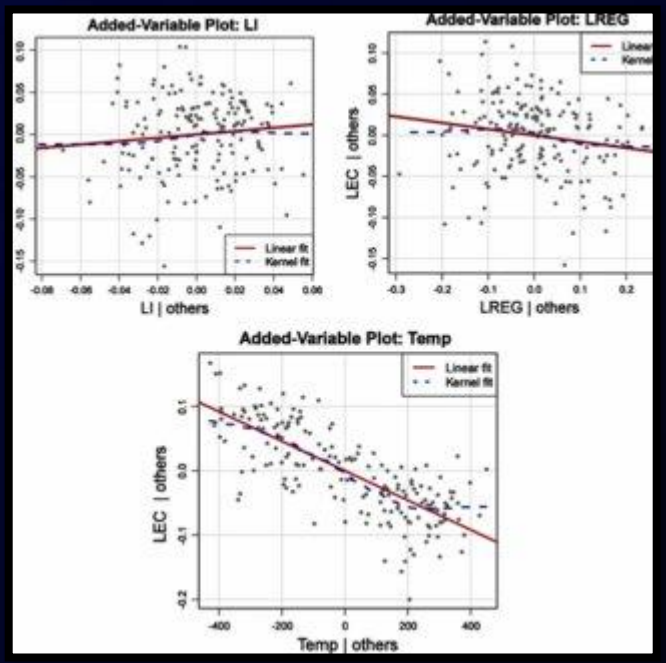
Sum-of-Squares Analysis,
Standardization, & Ridge Regression

Dr. Matthew N. Moore

Summary of MLR Regression Metrics/Concepts

Quick Summary

Concept	Symbol
Correlation Coefficients	$r_{x_k,y}$ and r_{x_j,x_k}
Coefficients of Determination	r^2 and r^2_{adj}
Model Coefficients	$\hat{\beta}_k \rightarrow \mathrm{SE}(\hat{\beta}_k) \rightarrow t^{\beta_k}_{stat}$
ANOVA Metrics	SSE, SSM, SST, MSE, MSM, MST, F_{stat}
Multicollinearity Metrics	V_k and K_k
Leverages	$H \rightarrow l_k \rightarrow \hat{\varepsilon}^s_k$
Added Variable Plots	$\hat{\beta}^{AV}_{1,k}$
Partial Correlations	$r_{(x_k,y)-x_j} = \dfrac{\left(r_{x_k,y}\right) - \left(r_{x_k,x_j}\right)\left(r_{x_j,y}\right)}{\sqrt{\left(1 - r^2_{x_k,x_j}\right)\left(1 - r^2_{x_j,y}\right)}}$



Sum of Squares Analysis

Type I – Sequential Sum of Squares

Added variable plots and partial correlations can tell you which variables are truly contributing to your model; one can also use sum of squares analysis to determine the percentage of effect those variables have for predictability in your model.

```
mydata = read.csv('CompanyPayroll_9.25.csv')
source('mymlr.R')

Y = mydata$Efficiency_Factor_Obs

X1 = mydata$Age
X2 = mydata$Prior_Experience
X3 = mydata$Hourly_Wage
X4 = mydata$Hours_Worked

Xk = cbind(X1, X2, X3, X4)
M = mymlr(Xk, Y)
```



```
> M$RMSE
[1] 1.676095
> M$SSE
[1] 227.5528
> M$SSM
[1] 68.18945
```

```
M_1 = mymlr(X1, Y)
M_1$SSM

M_12 = mymlr(cbind(X1, X2), Y)
M_12$SSM

M_123 = mymlr(cbind(X1, X2, X3), Y)
M_123$SSM

M_full = mymlr(Xk, Y)
M_full$SSM
```



Variables	SSM	ΔSSM
∅	0	0
1	17.9770	17.9770
1,2	18.2066	0.2296
1,2,3	47.6043	**29.3977**
1,2,3,4	68.1895	20.5852

Sum of Squares Analysis

Type I – Sequential Sum of Squares

Does the order for which you introduce variables matter?

```
> mymlr_sss(Xk, Y, ord=c(1,2,3,4))
  Variables      SSM Delta_SSM
1         Ø  0.00000  0.000000
2         1 17.97701 17.977011
3       1,2 18.20657  0.229563
4     1,2,3 47.60427 29.397697
5   1,2,3,4 68.18945 20.585181
```

```
> mymlr_sss(Xk, Y, ord=c(1,3,4,2))
  Variables      SSM  Delta_SSM
1         Ø  0.00000  0.0000000
2         1 17.97701 17.9770108
3       1,3 45.92939 27.9523762
4     1,3,4 67.56860 21.6392156
5   1,3,4,2 68.18945  0.6208498
```

```
> mymlr_sss(Xk, Y, ord=c(2,4,3,1))
  Variables      SSM  Delta_SSM
1         Ø  0.000000  0.0000000
2         2  4.559038  4.5590381
3       2,4 53.333593 48.7745546
4     2,4,3 67.631608 14.2980158
5   2,4,3,1 68.189452  0.5578439
```

```
> mymlr_sss(Xk, Y, ord=c(4,3,2,1))
  Variables      SSM  Delta_SSM
1         Ø  0.00000  0.0000000
2         4 53.31425 53.3142469
3       4,3 67.25049 13.9362480
4     4,3,2 67.63161  0.3811135
5   4,3,2,1 68.18945  0.5578439
```

```
> mymlr_sss(Xk, Y, ord=c(2,1,4,3))
  Variables      SSM Delta_SSM
1         Ø  0.000000  0.000000
2         2  4.559038  4.559038
3       2,1 18.206574 13.647536
4     2,1,4 54.271849 36.065275
5   2,1,4,3 68.189452 13.917603
```

```
> mymlr_sss(Xk, Y, ord=c(3,1,2,4))
  Variables      SSM Delta_SSM
1         Ø  0.00000  0.000000
2         3 33.66033 33.660329
3       3,1 45.92939 12.269058
4     3,1,2 47.60427  1.674884
5   3,1,2,4 68.18945 20.585181
```

It's evident that the order of the predictors matter and hence give different conclusions.

Number of hierarchies possible is $p!$

From all of them, one could do a majority vote for the results.

Sum of Squares Analysis

Type II – Full/Reduced (Drop 1) Sum-of-Squares

Full/reduced model comparisons consider the full model with all p predictors and analyze the change in SSM when you remove the predictor of focus. Instead of $p!$ calculations (like in Type I), this approach only has p calculations.

```
> mymlr_drop1(Xk, Y)
  Variable SSM_full SSM_minus_j  Delta_SSM Percentage_Change
1       X1 68.18945    67.63161  0.5578439       0.008180795
2       X2 68.18945    67.56860  0.6208498       0.009104778
3       X3 68.18945    54.27185 13.9176033       0.204101996
4       X4 68.18945    47.60427 20.5851812       0.301882190
```

Interpretations of both types:

- Type I: What does X_j explain after the predictors before it have entered, but before the predictors after it are allowed in?
  - Usually gets a "who gets credit first" interpretation; bad in the presence of multicollinearity.

- Type II: How much does X_j improve prediction once all other predictors are already in the model?

Q) What if there is interaction of any type of degree?

Sum of Squares Analysis

Type III – Generalized Full/Reduced (Drop 1) Sum-of-Squares

One can add interaction and polynomial terms into the model to address effects. Suppose now that your full model (as an example with $p = 4$) is:

$$Y \sim (X_1, X_2, X_3, X_4) \cup (X_1X_2, X_2X_4, X_1X_3X_4) \cup (X_1^2, X_2^2, X_2^3, X_4^2)$$

```
source('mymlr_drop1gen.R')

G = list(X1 = c(1,5,7,8),
         X2 = c(2,5,6,9),
         X3 = c(3,7,10),
         X4 = c(4,6,7,11))
Xkfull = cbind(X1, X2, X3, X4,
               X1*X2, X2*X4, X1*X3*X4,
               X1*X1, X2*X2, X2*X2*X2, X4*X4)
```

```
> mymlr_drop1(Xk, Y)
  Variable SSM_full SSM_minus_j  Delta_SSM Percentage_Change
1       X1 68.18945    67.63161  0.5578439        0.008180795
2       X2 68.18945    67.56860  0.6208498        0.009104778
3       X3 68.18945    54.27185 13.9176033        0.204101996
4       X4 68.18945    47.60427 20.5851812        0.301882190
```

```
> mymlr_drop1gen(Xkfull, Y, G)
  Variable_Group SSM_full SSM_minus_group Delta_SSM Percent_Change
1             X1 91.58201        75.83948  15.74252       0.1718954
2             X2 91.58201        81.57714  10.00487       0.1092449
3             X3 91.58201        80.54778  11.03422       0.1204846
4             X4 91.58201        62.14662  29.43538       0.3214101
```

Note that this method of sum-of-squares typically has heavy structural multicollinearity, likely making $\det(X^T X) \approx 0 \leq \varepsilon_{mac}$.

Predictor Standardization

What is it and why should we care?

We say that X_j is standardized (to Z_j) if

$$Z_j := \frac{X_j - \mathbb{E}(X_j)}{\sqrt{\mathbb{V}(X_j)}}$$

```
Zk = scale(Xk)

Mx = mymlr(Xk, Y)
Mz = mymlr(Zk, Y)
```

Q) Does this change the model in any way(s)?

```
> cbind(Mx$bhat, Mz$bhat)
            [,1]        [,2]
   -0.15747099  3.18148148
X1 -0.02224049 -0.11827439
X2  0.15808971  0.09783201
X3  0.07691795  0.44819619
X4  0.06834144  0.72806378
```

```
> c(Mx$Fstat, Mx$RMSE, Mx$mci)
                          X1       X2       X3       X4
5.693623 1.676095 1.416379 1.110536 1.074561 1.435281
> c(Mz$Fstat, Mz$RMSE, Mz$mci)
                          X1       X2       X3       X4
5.693623 1.676095 1.416379 1.110536 1.074561 1.435281
```

```
> X = cbind(1, Xk)
> Z = cbind(1, Zk)
> sqrt(abs(det(t(X)%*%X)))
[1] 7362059
> sqrt(abs(det(t(Z)%*%Z)))
[1] 36037.1
```

Standardization of predictors helps the stability of the model, but it doesn't affect the predictability of the coefficients.

***Note**: If you want to predict with model M_z, you must standardize your predict-with points with respect to the variables for which you built your model with.*

Ridge Regression

What is it and why should we care?

An alternative to least-squares regression:

$$\text{Minimize: } f(\hat{\beta}) = \left|\left|Y - X\,\hat{\beta}\right|\right|_2^2 \quad \rightarrow \quad \hat{\beta} = [X^T X]^{-1} X^T Y$$

is "ridge regression", which seeks to optimize over the functional:

$$\text{Minimize: } f(\hat{\beta}) = \left|\left|Y - X\,\hat{\beta}\right|\right|_2^2 + \lambda \cdot \left|\left|\hat{\beta}\right|\right|_2^2$$

where $\lambda \in \mathbb{R}_{\geq 0}$. Note that least-squares is a special case of ridge regression ($\lambda = 0$).

One can solve $\frac{\partial}{\partial \hat{\beta}}\left[(Y - X\,\hat{\beta})^T (Y - X\,\hat{\beta}) + \lambda \cdot \hat{\beta}^T \hat{\beta}\right]$ to find that:

$$\hat{\beta}_r = [X^T X + \lambda \cdot I]^{-1} X^T Y$$

```
mydata = read.csv('CompanyPayroll_9.25.csv')
source('mymlr.R')

Y = mydata$Efficiency_Factor_Obs

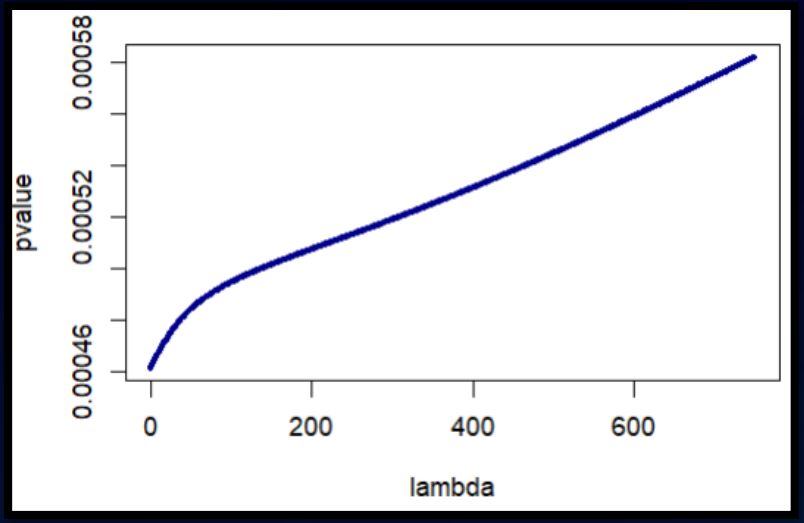
X1 = mydata$Age
X2 = mydata$Prior_Experience
X3 = mydata$Hourly_Wage
X4 = mydata$Hours_Worked
Xk = cbind(X1, X2, X3, X4)
```

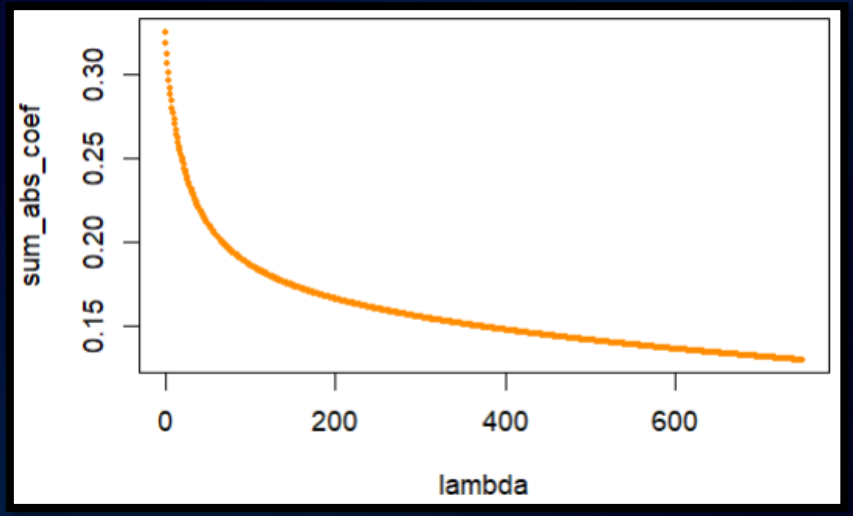
```
M = mymlr(Xk, Y, lam=0)
M_rdg = mymlr(Xk, Y, lam=50)
```

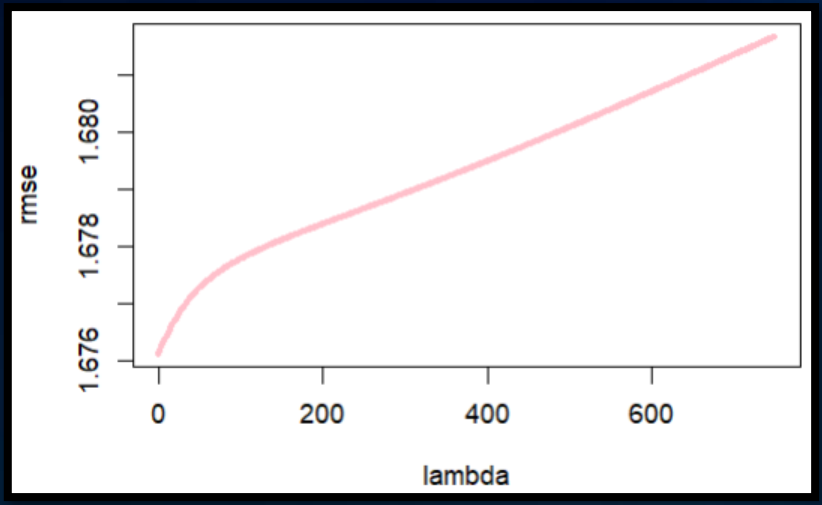
```
> c(M$RMSE, M_rdg$RMSE)
[1] 1.676095 1.676787
> rbind(t(M$bhat), t(M_rdg$bhat))
                            X1          X2          X3          X4
[1,] -0.157470985 -0.02224049 0.15808971 0.07691795 0.06834144
[2,] -0.007077778 -0.02143570 0.07645847 0.07311845 0.06947275
```

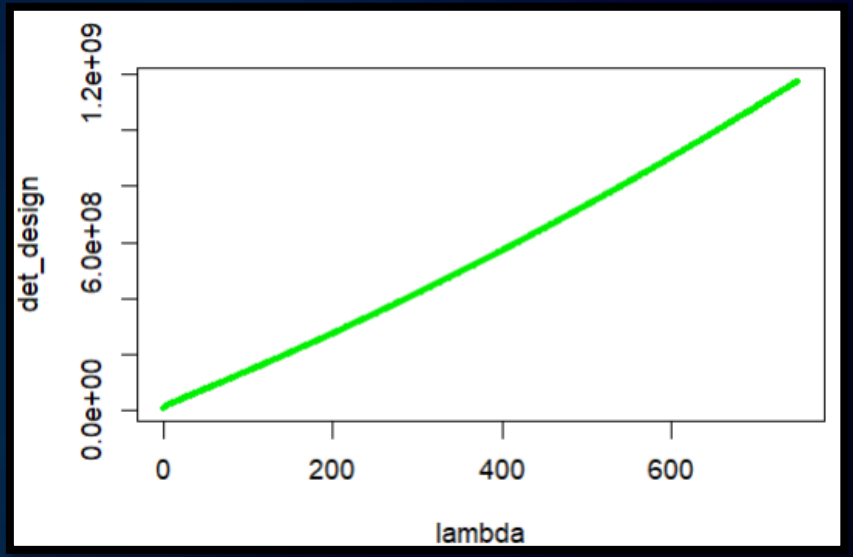
Ridge Regression

What is the effect of λ on the model?







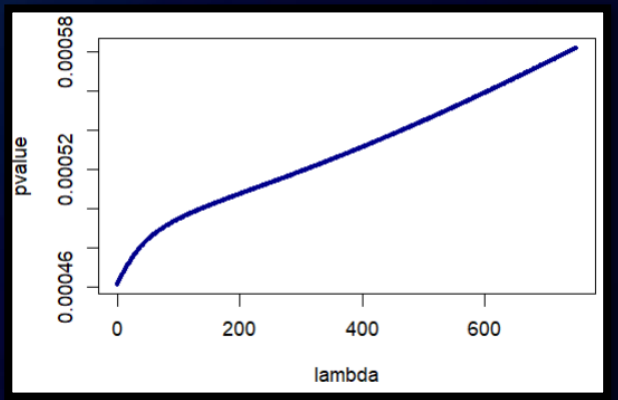


Ridge regression hence prioritizes stability of the regression model over the significance (of the predictors) for the regression model.

Ridge Regression

How to choose a reasonably ridge parameter?

Suppose model significance is decided (completely) by the p-value test statistic (there are SEVERAL things that must be satisfied for this to be valid), with significant decided by some α_{crit}.




```
plot(lms, mypval, pch=16, col='darkblue', cex=0.5,
     xlab = 'lambda', ylab='pvalue')

acrit = 0.05
abline(h = acrit, lty=2, col='red')

lcrit = which.min(abs(mypval-acrit))
abline(v=lcrit, lty=2, col='purple')
```

